In a ripple tank, circular water waves are produced by two vibrators S_1 and S_2 of the same frequency vibrating in phase. Their separation is 3.5λ, where λ is the wavelength of the waves. Figure 6.1 shows the two circular waves propagating on the water surface at a certain moment. Line L is a line connecting all points P which have path difference $S_1P - S_2P = 0$.

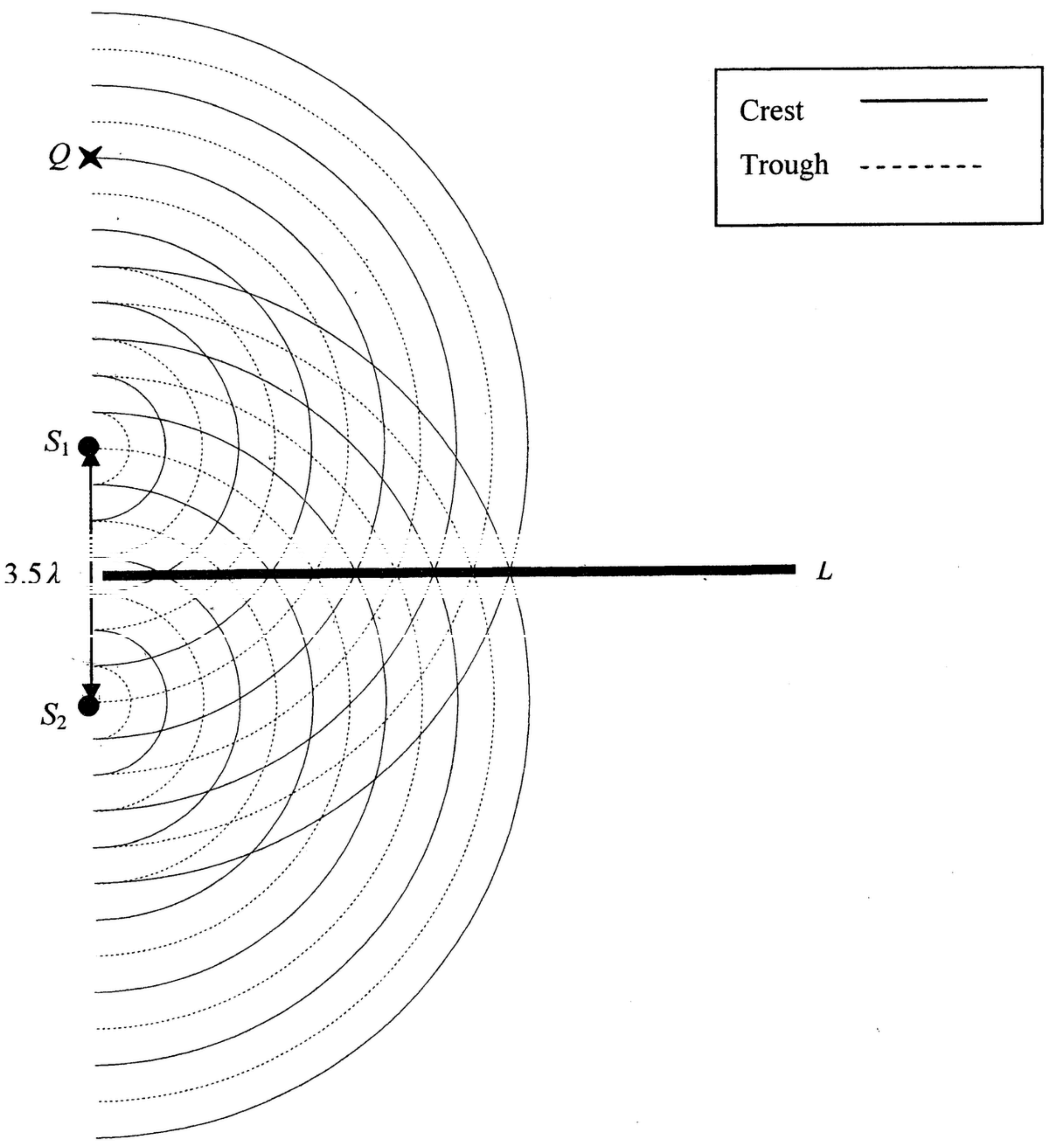

Figure 6.1

(a) Draw and label a line in Figure 6.1 connecting all points P which have path difference

(i) $S_1P - S_2P = \lambda$ (label it as L_1)

(ii) $S_1P - S_2P = \frac{3}{2}\lambda$ (label it as L_2)

What would happen to L_1 and L_2 if the separation between S_1 and S_2 is reduced slightly ? (3 marks)

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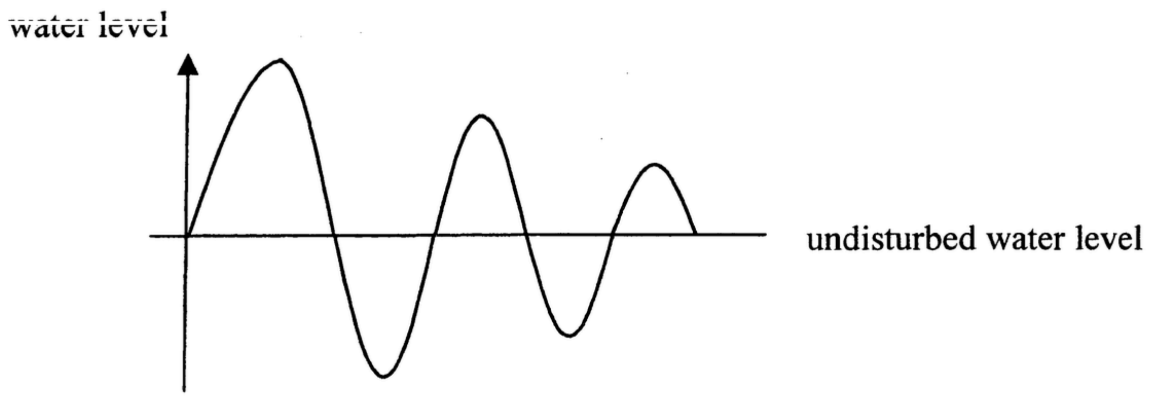
Answers written in the margins will not be marked.

(b) Figure 6.2 shows the profile of the water level along line L at a certain instant. Sketch on the same figure the profile at a time $\frac{T}{2}$ later, where T is the period of the water waves. (1 mark)

Figure 6.2



(c) Q is a point on the line joining S_1 and S_2 as shown in Figure 6.1. State the kind of interference that occurs at Q and give a reason for this occurrence. (2 marks)

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*(d) A similar double-slit set-up is used for the demonstration of the interference of light in which the separation between slits S_1 and S_2 is 0.5 mm and the screen is at 2.5 m from the slits. Calculate the average separation between adjacent bright fringes on the screen for a monochromatic light of wavelength 550 nm. (2 marks)

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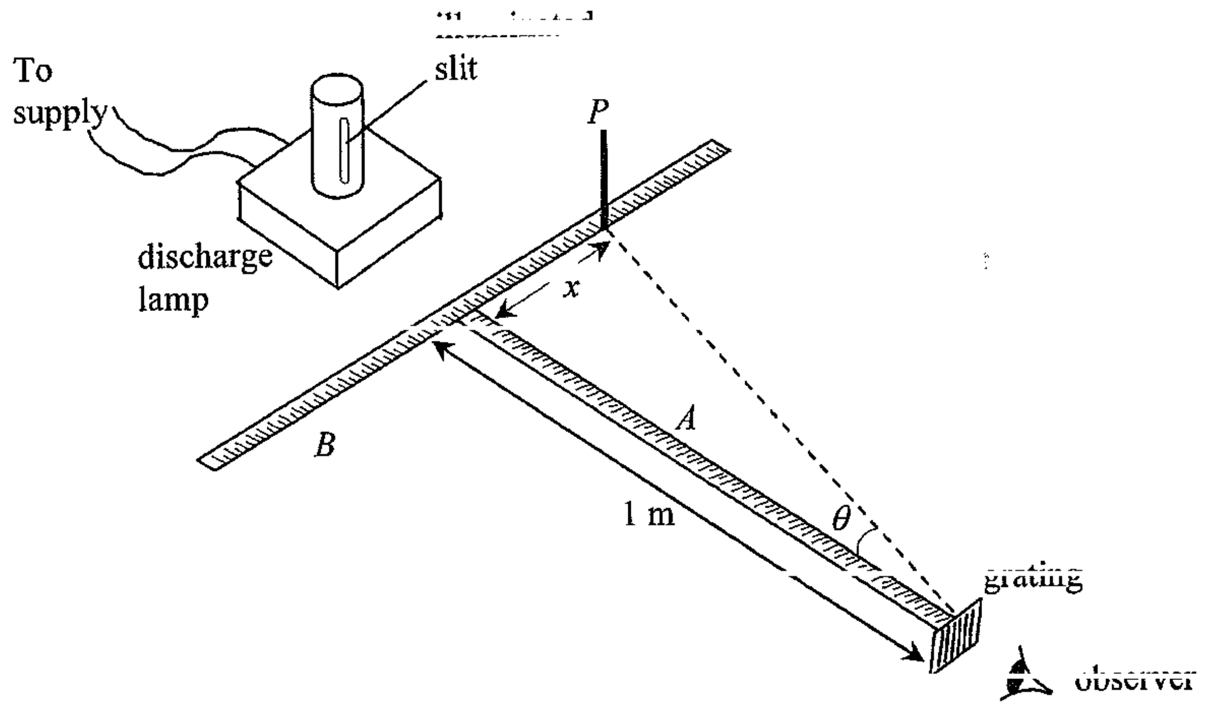
Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

7. Figure 7.1 shows an experimental set-up to determine the wavelength of monochromatic light emitted from the vertical narrow slit of a discharge lamp. *A*, *B* are two mutually perpendicular metre rules on the bench with rule *A* pointing towards the lamp. A diffraction grating with vertical lines is placed at one end of rule *A*. A vertically mounted pin *P* is moved along rule *B* until the pin is in line with the diffracted image of the second-order to the observer. The corresponding distance *x* is measured for finding the diffraction angle θ.

Figure 7.1



The grating has 300 lines per mm and *x* is found to be 0.38 m for the **second-order** image.

(a) (i) Calculate the diffraction angle θ. (1 mark)

*(ii) Hence find the wavelength of the light from the lamp. (3 marks)

(a) (iii) Give **ONE** advantage of measuring the position of the second-order image instead of the first-order one. (1 mark)

(b) In the experiment, the illuminated slit may not be well aligned along metre rule *A*. Suggest one way to reduce this error. (2 marks)

7. *(a)

Figure 7.1

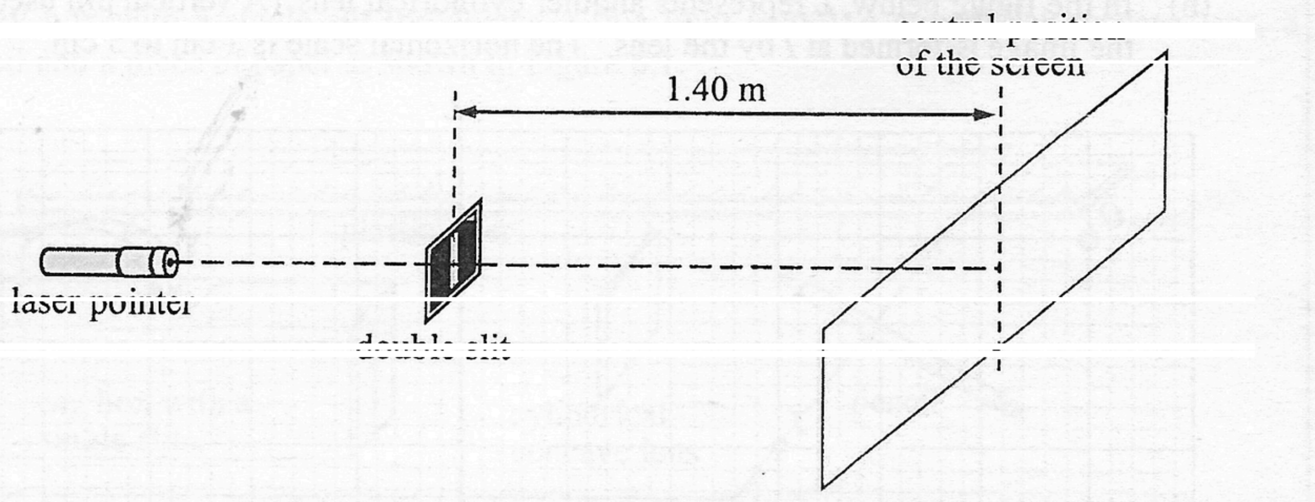


Figure 7.1 shows a set-up for measuring the wavelength λ of light emitted by a laser pointer. Several bright dots of average separation about 2 mm can be seen on the screen.

(i) For the same set of apparatus, suggest a way to increase the average separation between the bright dots on the screen. (1 mark)

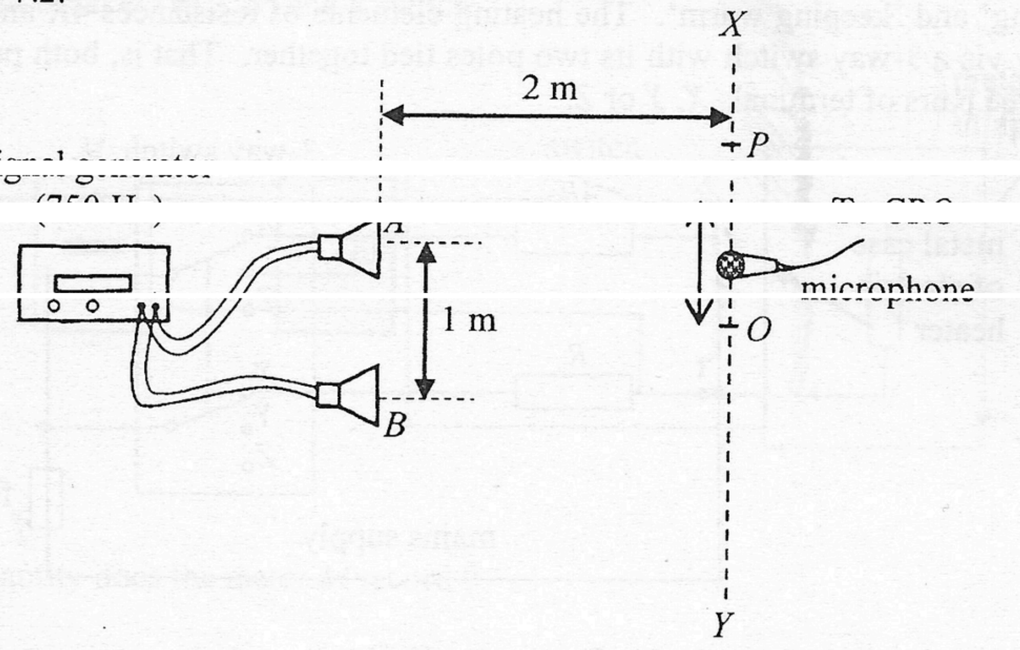
The double slit is now replaced by a diffraction grating with 400 lines per mm.

(ii) Briefly explain why the accuracy of the experiment can be improved. (1 mark)

(iii) Only five bright dots are observed on the screen such that the separation between the 1st and 5th dots is 1.56 m. Find λ. (3 marks)

(b) To measure the speed of sound in air, a student connects two loudspeakers, A and B, to a signal generator as shown in Figure 7.2.

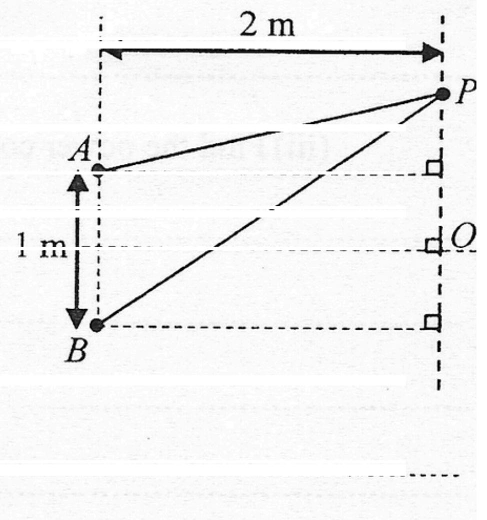
Figure 7.2



The separation of A and B is 1 m. A microphone is used to pick up the sound along the line XY at a distance of 2 m from the loudspeakers. The central maximum is at point O while the next maximum is at point P.

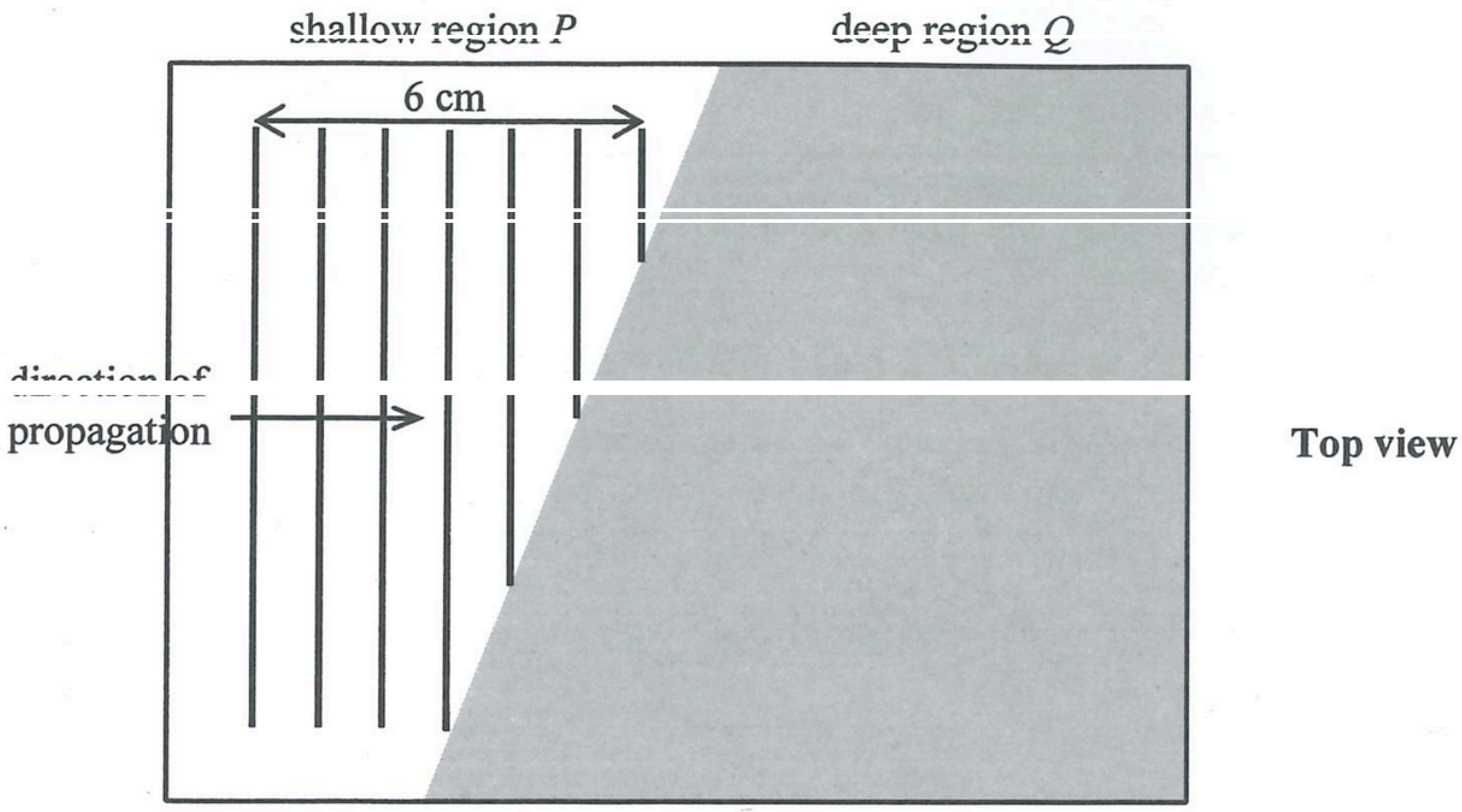
(i) With reference to the above settings, use the fringe separation equation $\Delta y = \frac{\lambda D}{a}$ in double-slit interference to find the wavelength λ of sound is not accurate. Explain briefly. (1 mark)

(ii) The distance between O and P is found to be 1 m when the signal generator is set at 750 Hz. By considering the path difference $PB - PA$, use the results of the experiment to find the speed of sound in air. (3 marks)



5. A ripple tank has a shallow region P and a deep region Q. Straight water wave of frequency 10 Hz is travelling in the shallow region as shown in Figure 5.1 when viewed from above.

Figure 5.1



(a) The separation between seven crests in the shallow region is found to be 6 cm as shown.

(i) Find the wavelength of the wave in the shallow region. (1 mark)

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(ii) What is the wave speed in the shallow region ? (1 mark)

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(b) The water wave then propagates into the deep region where the wavelength of the wave is double that in the shallow region.

(i) State the frequency of the water wave in the deep region. (1 mark)

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(ii) On Figure 5.1, sketch the wave pattern in the deep region. (2 marks)

(iii) Name the phenomenon occurred across the boundary and explain its cause. (2 marks)

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Answers written in the margins will not be marked.

*6. After removing the protective coating and the reflective layer of a compact disc (CD), the CD becomes a diffraction grating with evenly spaced parallel slits (Figure 6.1(a)). The set-up in Figure 6.1(b) employs a laser beam which can be used to measure the slit separation d.

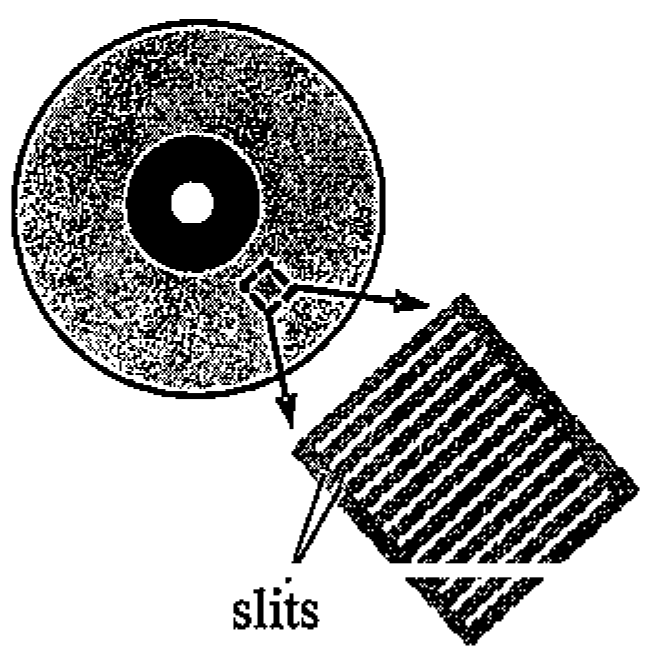

Figure 6.1(a)

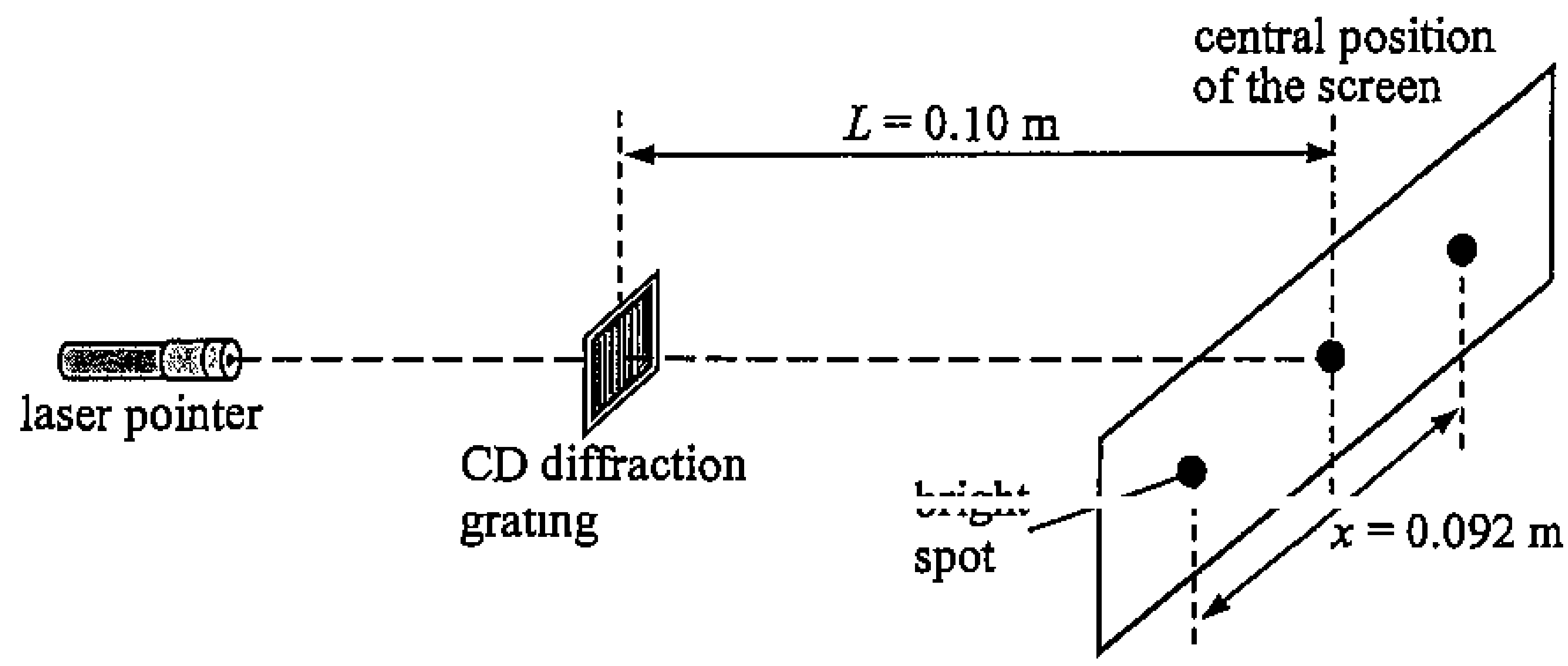

Figure 6.1(b)

Given that the separation between the CD diffraction grating and the screen, L, is 0.10 m, the separation between the first-order maxima, x, is 0.092 m and the wavelength λ of the red laser light beam used is 650 nm.

(a) Estimate d. (3 marks)

(b) State and explain the change in x if green laser light is used. (2 marks)

Answers written in the margins will not be marked